IBM Data Science Capstone Project

SpaceX Falcon 9 Launch Analysis

Predicting First-Stage Landing Success

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IBM Data Science Professional Certificate

Tools: Python, SQL, Pandas, Scikit-learn, Folium and Plotly

Executive Summary

- Analyzed historical SpaceX Falcon 9 launch data.
- Explored launch outcomes, payload characteristics and launch-site patterns.
- Used SQL queries and interactive geospatial visualizations to investigate the data.
- Developed a machine-learning classification workflow to study landing outcomes.
- Combined data analysis and visualization to support evidence-based conclusions.

Project objective: Investigate factors associated with first-stage landing success and assess whether historical launch data can help predict outcomes.

Introduction and Business Problem

- SpaceX uses reusable launch vehicles to reduce the cost of spaceflight.
- The Falcon 9 first stage can return to Earth and be recovered for potential reuse.
- Landing outcomes are influenced by launch and mission characteristics.
- Historical launch data can be analyzed to identify patterns in landing outcomes.

Business question: Can historical launch information help estimate the probability of a successful first-stage landing?

Data Collection and Data Wrangling

- Attempted to collect historical launch data from the SpaceX API.
- Encountered connection errors during the API request and used available historical course datasets for analysis.
- Inspected the data structure, column types and missing values.
- Converted launch dates into datetime format.
- Examined duplicate records and handled missing landing-pad information.
- Prepared features and the target variable for subsequent analysis.

Dataset used in Notebook 3: 90 rows and 17 columns.

Tools: Python, Requests and Pandas.

Exploratory Data Analysis (EDA)

The analysis examined relationships between launch characteristics and landing outcomes.

- Compared outcomes across launch sites.
- Investigated flight number and landing outcome.
- Examined payload mass distributions.
- Used visualizations to identify possible trends and differences.
- Explored whether launch characteristics may be associated with landing success.

Tools: Matplotlib, Seaborn and Pandas.

Data Analysis Using SQL

SQL queries were used to explore the SpaceX launch dataset.

- Retrieved launch records and examined relevant fields.
- Grouped records by launch site and outcome.
- Investigated payload mass and other launch characteristics.
- Used aggregation and filtering to summarize the data.
- Compared SQL-based summaries with findings from the Python analysis.

Tool: SQLite.

Key findings: Insert two or three verified results from Notebook 5.

Geospatial Analysis Using Folium

- Created an interactive map using launch-site coordinates.
- Displayed launch locations with map markers.
- Enabled geographic exploration of the launch sites.
- Connected spatial information with the broader launch-data analysis.

Tool: Folium.

Purpose: Make the geographic distribution of launch sites easier to explore and interpret.

Interactive Visualization with Plotly

- Developed interactive visualizations to explore launch outcomes.
- Examined payload mass and launch-site categories.
- Used interactive charts to support data exploration.
- Designed the analysis to make comparisons easier for stakeholders.

Tools: Plotly and Python.

Dashboard purpose: Allow users to explore launch data and investigate patterns through interactive visual elements.

Predictive Analysis Methodology

- Prepared input features and the landing-outcome target.
- Split the data into training and testing sets.
- Built classification models to predict the target outcome.
- Used model evaluation methods to assess predictive performance.
- Compared model results to understand their strengths and limitations.

Tools: Scikit-learn, Pandas and Python.

Important: The target label and feature rows must be correctly aligned before interpreting model performance.

Predictive Model Results

Model evaluated: [Insert model name]

- Test accuracy: [Insert verified value]
- Precision: [Insert verified value]
- Recall: [Insert verified value]
- F1-score: [Insert verified value]

Interpretation: Discuss how well the model identifies successful and unsuccessful landing outcomes, including the impact of any misclassifications.

Limitation: Results depend on data quality, target definition, feature selection and the size of the test dataset.

Conclusions and Key Insights

- Integrated data collection, wrangling, SQL, visualization and machine learning into one analysis workflow.
- Used exploratory analysis to investigate potential relationships between launch characteristics and landing outcomes.
- Used interactive mapping and charts to make the dataset easier to explore.
- Applied classification techniques to evaluate the feasibility of predicting landing outcomes.

Future improvements

- Expand the dataset and validate target labels.
- Compare additional classification algorithms and tune model parameters.
- Evaluate models with cross-validation and appropriate performance metrics.
- Incorporate additional mission and launch variables where available.

Overall conclusion: Historical launch data provides a foundation for studying landing outcomes, but predictive claims must be supported by validated model results.

Project Repository and References

GitHub repository

https://github.com/RAHULKR123456/IBM-SpaceX-Capstone

Project components

- Eight Jupyter notebooks
- Data collection and preparation
- EDA and SQL analysis
- Folium interactive map
- Predictive analysis
- Plotly visualizations

References

- IBM Skills Network SpaceX Capstone course datasets
- Historical SpaceX launch information from Wikipedia
- SpaceX API, where accessible

Thank you